

Capstone Project - NCCPSS Database Design

1. Problem Statement

The North Carolina Certified Peer Support Specialist Program manages certification for nearly 7,000 peers across North Carolina. Program data is currently stored across multiple Excel and Access files. This makes it difficult to combine data, track participants over time, ensure accuracy, and perform analysis.

Because there is no centralized database, the program faces challenges with reporting, longitudinal tracking, and decision making based on data. This limits the program's ability to identify trends, monitor progress, and improve coordination with partners and stakeholders.

The goal of this project is to design a centralized, accurate, and well structured database that supports certification management, reporting, and analysis for the program.

2. Background

The NCCPSS Program operates in partnership with the North Carolina Department of Health and Human Services and the Division of Mental Health, Developmental Disabilities, and Substance Use Services. The program oversees certification and ongoing program management for thousands of peer support specialists statewide.

Historical data has grown over time and is now spread across disconnected systems. This creates inefficiencies and increases the risk of data errors. A unified system would improve oversight, consistency, and long term program evaluation.

3. Stakeholders

UNC School of Social Work

North Carolina Certified Peer Support Specialist Program

North Carolina Department of Health and Human Services

Division of Mental Health, Developmental Disabilities, and Substance Use Services

Program staff, trainers, and partners

Certified peer support specialists

4. Team & Roles

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5. Data Plan

The NCCPSS data currently comes from multiple sources, including a Microsoft Access registry, a Drupal-based application and recertification system, and supplemental Excel files. Our data plan focuses on consolidating these sources into a single, well structured system that separates stable participant information from time based records such as certification, recertification, and employment outcomes. We developed a data dictionary and a proposed data structure to guide data cleaning, validation, and historical tracking. This approach supports longitudinal analysis, improves reporting accuracy, and allows for future scalability and automation.

Existing data sources include Access databases, application forms, workflows, and reporting outputs. The team will assess all current data sources to identify required fields, data quality issues, and reporting needs.

Data will be cleaned, standardized, and migrated into a new centralized database designed to support tracking over time and analysis.

Additional data sources include the PS Registry Access database and Drupal, which has been used for recertification data starting in 2024. Data from these systems may be exported to CSV or JSON formats for analysis and integration. Some data cleaning and standardization will be required due to manual data entry and inconsistent field naming.

6. Approach & Methodology

The team will first assess current systems and data sources to define requirements. A database structure will then be designed to support consolidation, tracking, and reporting. Existing data will be migrated into the new system and validated for accuracy.

The system will be tested with program staff and refined based on feedback before broader implementation. As part of the database design process, the team will create a data dictionary to clearly define fields, data types, and relationships, as a formal data dictionary does not currently exist. The database schema will be designed to support certification history over time and allow for future automation of data updates.

7. Timeline & Milestones

Assess existing data and systems

Design database schema

Clean and migrate data

Test system with stakeholders

Refine and finalize database

Prepare final presentation and documentation

Dates to be updated as the project progresses

8. Deliverables & Presentations

Centralized database for the NCCPSS Program

Clean and documented data structure

Ability to track participants over time

Standardized reporting and workflows

Visualizations or dashboards for program insights

Final presentation and project documentation

9. Risks

Some potential risks include inconsistencies between the Access and Drupal data, limited time within the semester, and possible changes in stakeholder needs. We also need to consider university infrastructure policies and any Microsoft licensing or funding requirements tied to long-term use of Fabric. Although the dashboards will present aggregate and anonymized data, final implementation may still depend on institutional approval and technical validation.

10. Team Literature Review

Our team reviewed academic research, government reports, and industry case studies related to peer support programs, legacy data systems, data modernization, and reporting and analytics. The literature shows that fragmented systems like Excel and Access limit reporting, historical tracking, and decision making, especially for public health and workforce programs. Prior work emphasizes the importance of centralized data structures, longitudinal tracking, and reliable reporting to evaluate outcomes such as employment and retention. These findings directly support the goals of the NCCPSS project and inform our approach to database design and analysis.

References

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